As a general rule, the size of the probe particle that we use sets a lower limit to the length scale to which we are sensitive. To get a feel for what this important statement means, imagine that Slim and Jim decide to get some culture by enrolling in a drawing class. As the semester progresses, Jim becomes increasingly irritated by Slim's growing proficiency as an artist and challenges him to an unusual contest. He proposes that they each take a peach pit, secure it in a vise, and draw their most accurate "still life" renditions. The unusual feature of Jim's challenge is that neither he nor Slim will be allowed to look at the peach pits. Instead, each is allowed to learn about the size, shape, and features of his peach pit only by shooting things (other than photons!) at the pit and observing how they are deflected, as illustrated in Figure 6.4. Unbeknownst to Slim, Jim fills Slim's "shooter" with marbles (as in Figure 6.4(a)) but fills his own shooter with far smaller five-millimeter plastic pellets (as in Figure 6.4(b)). They both turn on their shooters, and the competition begins.

After a while, the best drawing Slim can come up with is that in Figure 6.4(a). By observing the trajectories of the deflected marbles he was able to learn that the pit is a small, hard-surfaced mass. But that's all he could learn. Marbles are just too large to be sensitive to the finer corrugated structure of the peach pit. When Slim takes a look at Jim's drawing (Figure 6.4(b)), he is surprised to see that he has been outdone. A momentary glance at Jim's shooter, though, reveals the trick: The smaller probe particles used by Jim are fine enough to have their angle of deflection affected by some of the largest features adorning the pit's surface. And so, by shooting many five-millimeter pellets at the pit and observing their deflected trajectories, Jim was able to draw a more detailed image. Slim, not to be outdone, goes back to his shooter, fills it with even smaller probe particles—half-millimeter pellets—that are tiny enough to enter and hence be deflected by the finest corrugations on the pit's surface. By observing how these impinging probe particles are deflected, he is able to draw the winning rendition shown in Figure 6.4(c).

The lesson taught by this little competition is clear: Useful probe particles cannot be substantially larger than the physical features being examined; otherwise, they will be insensitive to the structures of interest.